THE UNIVERSITY OF HONG KONG
HKU BUSINESS SCHOOL
2021-2022 2nd Semester Examination I

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Finance: FINA2322
Derivatives
Sub-class E, F, G: Dr Yang Liu

Mar 22, 2022 7:00-8:30 p.m.

There are two parts in the exam.

Part I. 7 Multiple Choice Questions (5 point each, 35 points in total)

Part II. 6 Short Questions (65 points in total)

The total points are 100.

Total Pages of the Exam Paper (including this cover page): 6

- **You have to write down your Name and UID on the Answer Script.**
- **Make sure you show the steps and equations to score full credit.**
- **Leave 4 d.p. for the answers.**

Academic Integrity Statement: By taking this examination, students agreed to the following Academic Integrity Statement.

A. I acknowledge that University examinations require all students to respect the highest standards of academic integrity. For the examination I am about to take, I make the following pledge:

1. All the work will be my own, and I will not plagiarize from anyone else;

2. I will not obtain or seek to obtain an unfair advantage by communicating or attempting to communicate with any other person during the examination; neither will I give or attempt to give assistance to another student in taking the examination;

B. I understand that students who are suspected of violating this pledge are liable to be referred to the Disciplinary Committee, and maybe subject to disciplinary action such as suspension of studies or expulsion from the University.

Case 1 Hong Kong Motor Bus Company

Hong Kong Motor Bus Company (HMB) is your client and operates most bus routes in Hong Kong. With the war in Ukraine, the oil price has been very volatile and appears to have a upward trend recently. HMB has come to you for advice in risk management.

First of all, HMB buys oil in operation. They are considering whether or not to hedge their oil price risk. You explained to them what the advantages of hedging are and also explained reasons not to hedge.

MC Question 1 (5 points)
Which of the following is a reason that HMB wants to hedge risk?

A) Monitoring costs
B) Bankruptcy costs
C) Transaction costs on oil derivatives
D) Costs in hiring experts to monitor the positions

MC Question 2 (5 points)
HMB has an inherent ___(1)______. They could hedge his position by ___(2)___.

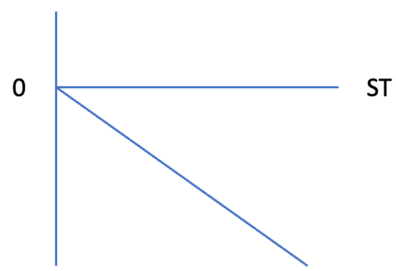
	(1)	(2)
A)	Short position in the oil price.	Short a collar on oil.
B)	Long position in the oil price.	Long a collar on oil.
C)	Short position in the oil price.	Short a call on oil.
D)	Long position in the oil price.	long a call on oil.

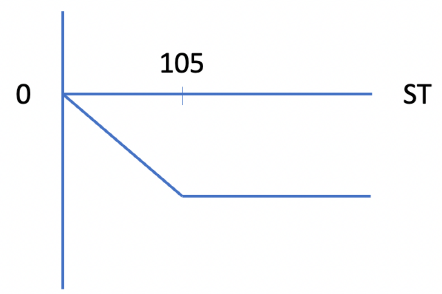
Short Question 1 (8 points)

HMB has scheduled to buy oil exactly 1 year from now (1 March). HMB is considering a strategy that would limit the maximum price they pay for oil (strategy 1). The exercise price in strategy 1 is US$105 per barrel.

(a) What is the unhedged cashflow per barrel? Draw the payoff diagram. (4 points)

(b) What is the hedged cashflow per barrel? Draw the payoff diagram. (4 points)





HMB found that strategy 1 appears to be costly and would like to pay a lower cost for the insurance strategy.

MC Question 3 (5 points)
Which of the following strategy can be a hedge and is less costly than the original strategy 1?

A) Short Collar with strike prices US$100 and US$105 per barrel
B) Short Forward
C) Long put with strike price of US$105 per barrel
D) Long call with strike price of US$100 per barrel

MC Question 4 (5 points)

HMB considers to use a 1-year forward contract to hedge his position. When the contract was entered, the forward price was US$110 per barrel. If 3-month later, the 1-year forward price is US$108 per barrel while the 9-month forward price is US$112 per barrel. What will be the value of HMB's forward position then?

A) Zero
B) Positive
C) Negative
D) Unknown

Your client is excited to learn about the forward contract. He asked about whether he can use the forward contract to predict the oil price in the future.

MC Question 5 (5 points)

HMB asked about the differences between the long forward contract on oil and a direct investment in oil. You have some thoughts when you explain to HMB about forwards.

1. When you take a long position in the forward contract, the initial value of the contract is zero.
2. When you buy a forward, you expect to earn compensation for the time value of money.
3. When you buy a forward, you expect to earn compensation for risk.
4. When there are transaction costs, it is impossible to earn arbitrage profit from forward.
5. Forward has less credit risk as compared to futures.

How many of the above statements are true?

A) 2
B) 3
C) 4
D) 5

Short Question 2 (16 points)

Eventually, HMB decides to use collar to hedge its risk exposure to oil price risk. (Strategy 2)

The following options expire in 1 year and the 1-year interest rate is 5% effectively. The oil price for today (March 1) is US$105 per barrel. HMB needs to have 8000 barrels 1-year later. Below is the relevant call and put premiums, with contract specifications: (you need to take into account the bid/ask price)

Strike Price	Call premium (USD)		Put Premium (USD)	
	Bid	Ask	Bid	Ask
100	17.4	17.6	7.53	7.73
110	12.58	12.78	12.22	12.42

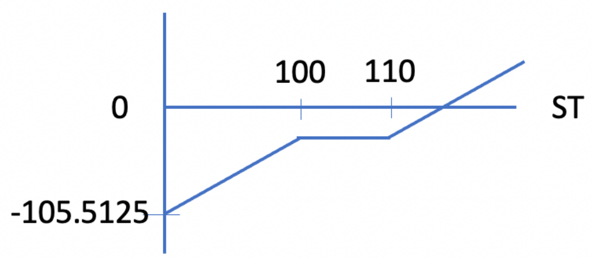
Contract Specifications	
Underlying asset	Oil
Contract unit	100 barrels
Price quotation	U.S. dollars and cents per barrel
Maturity	1 year

(a) What is the notional value of for each option contract? (3 points)
Notional value = 100*105 = $10500

(b) How many contracts should HMB buys? (2 point)
Long 80 calls, short 80 puts

(c) Find out the cost of strategy 2 **per barrel**. (3 points)
Cost = 12.78 – 7.53 = $5.25

(d) Draw the profit diagram of strategy 2 **per barrel**. (4 points)



(e) What is the oil price such that HMB can break-even on strategy 2? (4 points)

Break-even Price = 110 + 5.25*1.05 = 115.5125

MC Question 6 (5 points)

HMB expects to earn in total of HKD7.41 million (equivalent to USD950,000). Which of the following strategy cannot avoid a negative profit?

A) Long forward with forward price 110.25
B) Long call option with strike price 100
C) Long call option with strike price of 110
D) Long call option with strike price 110 and short put option with strike price 100

Case 2 Derivatives Asset Management

You are a fund manager in Derivatives Asset Management Fund (2322.HKU). You manage a portfolio consists of stocks and derivatives contracts. You profit mainly from arbitrage activities from mispricing and speculating on the movements and volatility of the market.

On a regular day, you will project your price expectation on the market. The current Hang Seng Index level is 21500. The risk-free interest rate is 0.3% p.a., continuously compounded. The dividend yield of HSI is expected to be 4% p.a., continuously compounded for the coming 3 months.

Below is your expectation of HSI level 3 months later:

Probability	Hang Seng Index level
0.1	19500
0.5	20000
0.2	21000
0.2	22000

In order to speculate in the market, you are considering the following strategies:

Strategy 1: Short call option with a strike price of 21400 and long put option with a strike price of 21000.
Strategy 2: Short forward on HSI with 3-month maturity.

All of the options used are with 3-month maturity. Below is the relevant call and put premiums (refer to the last trading price of the option, no need to take into account the bid/ask price):

CALL					JUN-22	PUT				
OI	Volume	IV	Bid/Ask	Last	Strike	Last	Bid/Ask	IV	Volume	OI
24	2	31.1%	-/-	1,285	20,600	-	-/-	-	-	42
68	1	28.1%	-/-	1,419	20,800	-	-/-	-	-	59
156	8	27.3%	387 /-	1,270	21,000	1,362	-/-	26.2%	3	665
26	2	27.2%	-/-	1,184	21,200	1,353	-/-	26.5%	6	411
66	2	27.8%	-/-	1,155	21,400	1,527	-/-	27.3%	1	403

Short Question 3 (22 points)

(a) What is the expected profit of strategy 1? (6 points)
Expected payoff of short call (K=21400) = 0.1*0 + 0.5*0 + 0.2*0 + 0.2*-600 = -120
Expected payoff of long put (K = 21000) = 0.1*1500 + 0.5*(1000) = 650
Expected profit = 650 – 120 – (1362 – 1155)*e^(0.3%/4) = 322.8447

(b) What is the 3-month forward price? (3 points)
Forward price = 21500*e^((0.3% - 4%)/4) = 21302.0420

(c) What is the expected profit of strategy 2? (3 points)
Expected price = 0.1*19500 + 0.5*20000 + 0.2*21000 + 0.2*22000 = 20550
Expected profit = 21302. 0420– 20550 = 752.0420

(d) You notice that the last trading price of put option with a strike price of 20800 is missing. Based on call premium in the table, what is the put premium implied by put-call parity? (4 points)
P = 1419 – e^(-0.3%/4)*(21302. 0420– 20800) = 917.3344

(e) You observed the forward price in the market is 21400 for a contract with 3 months to maturity. You planned to capture an arbitrage profit. Assume the size of a forward contract is 1 times the index level. Assume there are no transaction costs. What arbitrage would you undertake and what is the profit? (6 points)
Hint: you may use the table in answer sheet to help you consider the strategy.

Strategy	T=0	T=0.25
Short Forward	0	21400 – ST
Long HSI	-21500*e^(-4%/4) = - 21286.0714	ST
Borrow	21286.0714	-21286.0714e^(0.3%/4) = -21302.0420
Total	0	97.9580

Short Question 4 (6 points)

Suppose there are transaction costs. The lending rate and borrowing rate for 3-month are 0.3% p.a. and 0.4% p.a. respectively, continuously compounded. When you buy or sell the index, there is a transaction cost of 2% at t=0. There is also a fixed transaction cost for the forward contract of $20 per contract at t=0. There are no transaction costs 3-month later. If you arbitrage as in SQ3(e), what is the profit? (6 points)

Strategy	T=0	T=0.25
Short Forward	-20	21400 – ST
Long HSI	-21500*e^(-4%/4)*1.02 = - 21711.7929	ST
Borrow	21731.7929	-21731.7929e^(0.4%/4) = -21753.5355
Total	0	-353.5355

Short Question 5 (8 points)

Instead of trading the forward contracts, you switched to trade futures contract because of its high liquidity. Today, the 3-month HSI futures price is 21600 and you have entered 6 futures contracts in short position. The size of the futures contract is 50 times index point. The futures position is marked to market weekly. The initial margin is 10% while the maintenance margin is 7%. The margin account earns an interest of 1% p.a. continuously compounded.

(a) What is the initial margin account balance? (3 points)
Initial margin = 10%*21600*50*6 = $648000

(b) If 1-week later, the HSI futures price rises to 23000, what is your margin account balance? Will you receive a margin call? (5 points)

New margin account balance = 648000*e^(1%/52) + (21600 – 23000)*50*6 = 228124.6274
Maintenance margin = 7%*23000*50*6 = $483000
Therefore you will receive a margin call.

Short Question 6 (5 points)

You are also responsible for trading currency. You observed that today the exchange rate between US dollar and euros is 1.11 $/euro. Supposed that the 6-month risk-free interest rate in US is 5% p.a. and the 6-month risk-free interest rate is 3% in terms of euro. What should be the fair price of a 6-month currency forward? (5 points)
Forward exchange rate = 1.11*1.025/1.015 = 1.1209 $ per euro

MC Question 7 (5 points)

Based on your answer in short question 6, what arbitrage can you take if the 6-month forward price is 1.3 $/euro?

A) Long the currency forward, borrow US dollar, lend euros
B) Long the currency forward, lend US dollar, borrow euros
C) Short the currency forward, borrow US dollar, lend euros
D) Short the currency forward, lend US dollar, borrow euros